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First Rigless Implementation of Novel Approach for Saturation Time Lapse Logging Using Resistivity in deviated and horizontal Producer Wells- A Case Study from UAE's Onshore Field

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Abstract

Monitoring saturation changes and fluid front movement in producing wells is crucial for reservoir management planning, and subsequent water shut-off operations. Current logging time-lapse procedures (Pulsed Neutron Sigma and CO) face challenging conditions such as acid washes, low fluid salinity, small depth of investigation, and hole deviation.

This paper presents a new approach to measuring Deep and Medium Resistivity in Horizontal production wells to calculate an accurate time lapse fluid saturation.

The project is aimed to revive wells inactive due to water loading. Methods involved diagnosing unknown water influx sources and evaluating various solutions to log through completions with small restriction. Procedures included assessing technologies, selecting an innovative compact resistivity tool, and planning rigless deployment via coiled tubing in memory mode. Implementation entailed simulating and executing operation in a completed well. The new approach uses a slim dual induction resistivity tool measuring from inside a specially designed slim collar of max OD 2.5 inches. Under rigless operation, the tool can be deployed by coiled tubing in deviated or horizontal active/inactive wells.

The novel rigless resistivity tool, deployed for the first time in a UAE well, yielded high-quality data, confirming elevated water saturation across the reservoir interval. The resistivity log indicated uniform saturation increase, eliminating fracture-driven water influx as the cause of water logging. Observations highlighted the tool's reliability, with successful rigless coiled tubing deployment, marking the first deep resistivity application in horizontal well. Data processing corroborated accuracy, with saturation profiles aligning with production trends. The trial showcased the tool's ability to bypass size constraints of conventional resistivity tools, delivering open-hole-quality data in open hole environments. The advantage over conventional pulse neutron-based saturation monitoring is that for this novel technology of time lapse resistivity, the interpretation is apple to apple comparison with baseline and does not involve any subjectivity.

This trial resolves a global industry challenge by enabling direct saturation monitoring in completed wells and optimizing reservoir management in mature fields. Its rigless approach minimizes costs and environmental impact, offering scalability for global operators. Future enhancements, such as real-time telemetry could further strengthen its role in enhancing production and remedial strategies across diverse reservoirs, redefining surveillance.

This paper introduces the first rigless application of a compact resistivity tool in a completed well, a novel solution for saturation monitoring for an old well. It addresses an industry gap by enabling direct resistivity logging, surpassing the limitations of pulsed neutron logging. The trial's success in diagnosing water logging reasons enhances reservoir management, offering scalable, cost-effective surveillance and production enhancement for mature fields, thus advancing petroleum industry knowledge and practices.

PROBLEM STATEMENT

Effective reservoir management in carbonate reservoirs requires precise monitoring of fluid saturation changes and fluid front movement to optimize production and plan water shut-off operations in producing wells. Pulse neutron logging (PNL) has been the industry standard for time-lapse saturation monitoring, valued for its ability to assess fluid properties through casing in mature fields. However, many wells in the region are open-hole completions, where PNL and C/O logging lose their casing-penetrating advantage and face significant challenges, particularly in wells stimulated with acid recently. Recent operations revealed a critical discrepancy: PNL-derived saturations indicated low water saturation, yet well exhibited high water production, as evidenced by a horizontal open-hole well that ceased production due to water loading. A slickline survey confirmed the horizontal section was water-filled, contrasting original resistivity log showing only 12% water saturation. In open-hole wells stimulated recently, PNL and C/O logging are hindered by residual acidizing effects and operational complexities. Lingering saline fluids, such as spent acid or calcium chloride byproducts, distort sigma and C/O measurements, inflating water saturation estimates. Low fluid salinity in certain reservoirs reduces neutron capture contrast, obscuring fluid identification. Acid-induced wormholes and vugs create heterogeneous porosity, rendering hydrogen index-based porosity estimate unreliable. Residual CO₂ from carbonate dissolution can mimic hydrocarbon gas, leading to false positives in C/O logs. The shallow 6–12-inch depth of investigation of PNL fails to capture deeper reservoir conditions beyond the acid-altered zone. High deviation in horizontal open-hole completions further complicates interpretation, with fluid segregation, cross-flow, and tool sticking risks contributing to subjective interpretations and saturation-production mismatches. These challenges necessitated a novel approach for saturation time-lapse logging, utilizing a compact, resistivity-based tool with deep and medium resistivity measurements, spectral gamma-ray, and flowmeter capabilities, deployed via small-diameter coiled tubing in highly deviated wells. This method directly measures formation resistivity, mitigating acid-related effects and enabling objective baseline comparisons to diagnose water influx sources and enhance reservoir management.

Challenges of Pulsed Neutron Logging in Acid-Stimulated Wells

Pulse neutron logging (PNL) and carbon-oxygen (C/O) logging are essential for time-lapse saturation monitoring in cased-hole wells, but significant limitations hinder their application in open-hole carbonate reservoirs.

1. In open-hole completions, the absence of casing exposes PNL to borehole environmental interferences, such as saline drilling mud or formation water influx, which introduce noise in sigma and C/O measurements, compromising data reliability. Residual drilling mud filtrate, rich in hydrogen, distorts hydrogen index-based porosity estimates, leading to inaccurate assessments of reservoir quality in heterogeneous carbonates.

2. Borehole rugosity and irregular wall conditions, common in open-hole settings, scatter neutron signals, further degrading log quality, particularly in high-porosity zones where signal distortion is pronounced. These environmental factors require extensive calibration, often rendering PNL data less dependable for precise fluid saturation analysis in open-hole wells.
3. High well deviation in horizontal open-hole wells exacerbates PNL's challenges. Gravitational fluid segregation leads to uneven distributions of oil, water, and gas within the borehole, skewing PNL measurements and complicating the interpretation of flow regimes.
4. Cross-flow between reservoir zones, driven by pressure differentials, misallocates production contributions in multi-zone completions, as PNL struggles to resolve dynamic fluid movements.
5. Tool sticking risks in rugose open-hole sections increase operational downtime, with slickline or wireline interventions often requiring extended periods, such as 12–18 hours, for pure logging operations.
6. PNL's shallow depth of investigation, typically 6–12 inches, restricts measurements to near-wellbore properties, missing deeper reservoir conditions critical for accurate saturation analysis in complex carbonate formations.
7. Formation-specific factors in carbonate reservoirs further impair PNL performance. Low fluid salinity, often below 20,000 ppm NaCl, reduces neutron capture contrast, making fluid differentiation between oil and water ambiguous. Clay minerals, such as illite or smectite, prevalent in these reservoirs, contain bound water that elevates sigma values, falsely suggesting higher water saturation. Mixed wettability on carbonate surfaces alters neutron interactions, complicating hydrocarbon identification and skewing sigma measurements.
8. In open-hole wells recently stimulated with acid, PNL and C/O logging face amplified challenges due to the intricate chemical, physical, and operational alterations induced by acidizing. Acid treatments, typically hydrochloric acid, dissolve carbonate minerals, producing high-salinity byproducts like calcium chloride that persist in the near-wellbore zone, significantly elevating sigma values and leading to inflated water saturation estimates. These saline residues create localized conductivity anomalies that disrupt neutron capture measurements, masking hydrocarbon signatures, especially in low-salinity reservoirs where the contrast between formation water and acid byproducts is pronounced. Acid-induced dissolution forms wormholes, vugs, and micro-fractures, creating heterogeneous porosity distributions that render hydrogen index-based porosity estimates unreliable, particularly in vuggy zones where secondary porosity complicates reservoir characterization. Residual CO₂ or solution gas from carbonate dissolution produces carbon signatures in C/O logs that mimic hydrocarbons, resulting in erroneous gas saturation readings. The acid-altered near-wellbore zone, which may extend 6–18 inches, dominates PNL's shallow 6–12-inch depth of investigation, preventing access to deeper, unaltered reservoir conditions essential for time-lapse saturation analysis. Acidizing also triggers secondary reactions, such as precipitation of calcium carbonate, gypsum, or other scales, which form uneven coatings on borehole walls, attenuating neutron signals and further degrading data quality. In highly deviated horizontal wells, acid-enhanced borehole rugosity and weakened formation integrity increase tool sticking risks, with slickline or wireline tools often encountering obstructions in acid-etched sections, prolonging interventions and escalating operational complexity. Acid stimulation can also exacerbate cross-flow by creating high-permeability channels, leading to uneven fluid distribution and further distorting PNL's ability to accurately allocate production contributions. Chemical interactions between acid residues and formation fluids, such as emulsification or asphaltene precipitation, introduce additional signal noise, particularly in reservoirs with heavy oil or complex fluid compositions. These acid-specific challenges were evident in a recent case involving a horizontal open-hole well, post a scale clean-out job, where a slickline survey indicated a water-filled section, contradicting a prior resistivity log of 12% water saturation, underscoring the need for a more robust logging approach to resolve saturation-production discrepancies in an acid-stimulated environment

These issues lead to saturation-production discrepancies, as observed in cases where PNL indicated high water saturation, due to recent acidization in the well as well as its shallow depth of investigation, necessitating alternative logging methods for reliable reservoir management.

Novel Approach for Saturation Time-Lapse Logging

To overcome the challenges of pulse neutron logging (PNL) in accurately detecting petrophysical changes in complex open-hole carbonate reservoirs, several contractors were consulted, and a novel resistivity-based, rigless logging solution proposed by one company was selected for implementation. This solution features a compact, multi-sensor tool deployed via small-diameter coiled tubing (1.5–2 inches), integrating deep and medium resistivity measurements, spectral gamma-ray, and flowmeter capabilities. Unlike PNL, which is hindered by borehole, formation, and acid-related interferences, this approach measures formation resistivity directly, providing reliable time-lapse saturation data to enhance reservoir management in open-hole environments. This novel solution addresses key PNL limitations. It counters borehole interferences, such as saline mud or rugosity, with deep resistivity measurements extending 24–48 inches into the formation¹, far beyond PNL's 6–12-inch range, ensuring accurate reservoir property assessment. Spectral gamma-ray data corrects for clay-bound water, improving fluid identification in heterogeneous carbonates. In highly deviated horizontal wells, flowmeter measurements clarify fluid segregation and cross-flow, while coiled tubing minimizes tool sticking, enhancing efficiency compared to PNL's wireline deployments. The resistivity-based approach overcomes low fluid salinity and mixed wettability issues, delivering robust saturation estimates where PNL's neutron capture contrast is unreliable. For acid-stimulated wells, it penetrates past saline byproducts, wormholes, and scale, capturing unaltered formation properties and pinpointing water influx with flowmeter data. This contractor-proposed solution provides a versatile alternative to PNL, improving saturation monitoring and supporting effective water shut-off strategies in carbonate reservoirs.

Methodology

The methodology for deploying the novel resistivity-based, rigless logging solution was designed to overcome the limitations of pulse neutron logging in open-hole carbonate reservoirs, enabling accurate time-lapse saturation monitoring. The approach encompassed well selection, pre-job planning, tool deployment, memory-mode data acquisition, and post-retrieval analysis, tailored to the complex conditions of a candidate well. Multiple contractors were consulted, and a compact, multi-sensor tool—integrating deep and medium resistivity measurements, spectral gamma-ray, and flowmeter capabilities, deployed via 1.5–2-inch coiled tubing—was selected for its ability to navigate highly deviated wells and address environmental interferences.

Technical Requirement	Resistivity Time Lapse (Propagation)	Pulsed Neutron Time Lapse (PN)
Horizontal Deployment	Coiled Tubing / Tractor	Coiled Tubing / Tractor
Effects of Acid Jobs	Not affected	Highly affected (altered porosity response)
Depth of Investigation	45 – 55 in	15 – 24 in
Logging Speed	24 ft/min (faster)	15 ft/min (slower due to counting statistics)
Radioactive HSE Measures	None	Yes – requires sources, HSE handling & permits
Sensitivity to Lithology	Minimal impact in clean formations	Strongly dependent on matrix & borehole conditions
Modelling Requirements	Simple Archie-based interpretation	Requires complex sigma or C/O capture modelling
Time Lapse Repeatability	High (stable resistivity baseline)	Variable (affected by fluids, borehole changes, statistics)
Data Processing Complexity	Moderate (Archie/rock physics models)	High (nuclear modelling, corrections, reference logs needed)

Well Selection and Current Status. The candidate well was a single oil producer with a 3 1/2-inch completion, equipped with a chemical injection sub for corrosion and scale management, and approximately 6,500 ft of open-hole section to maximize reservoir contact. Completed in 2015, the well produced oil with minimal water cut for seven years. Since 2023, a gradual increase in water cut has led to production cessation by mid-2024 due to water loading. A Rigless scale clean-out job was performed to clear tubing obstructions, ensuring accessibility for logging operations. The well's history of acid stimulation (approximately 18 months prior) and recent production challenges necessitated precise saturation data to guide water shut-off, Artificial lifting methodology or sidetracking strategies.

Pre-Job Planning. Historical well data, including completion records, production logs, and prior PNL results, were reviewed to establish a baseline for saturation changes. Petrophysical models predicted fluid front movement and potential water influx zones. The coiled tubing unit, configured with a 2-inch coil, was prepared for deployment in the highly deviated (up to 90°) open-hole section. The tool was calibrated on-site to account for low-salinity formation water and acid-induced formation alterations, ensuring reliable resistivity and flowmeter measurements.

Tool Deployment and Data Acquisition. The compact, multi-sensor tool (Figure 1) was deployed Rigless via coiled tubing to access the 6,500-ft open-hole section without rig intervention. The tool was run to the target depth, navigating borehole rugosity and acid-etched sections to avoid sticking. Operating in memory mode, it recorded deep and medium resistivity measurements (probing 45–55 inches into the formation), gamma-ray data for lithology characterization, and flowmeter data to identify fluid influx points. Logging was conducted at a controlled speed to ensure high-resolution data capture, with the tool retrieved post-run for data download.

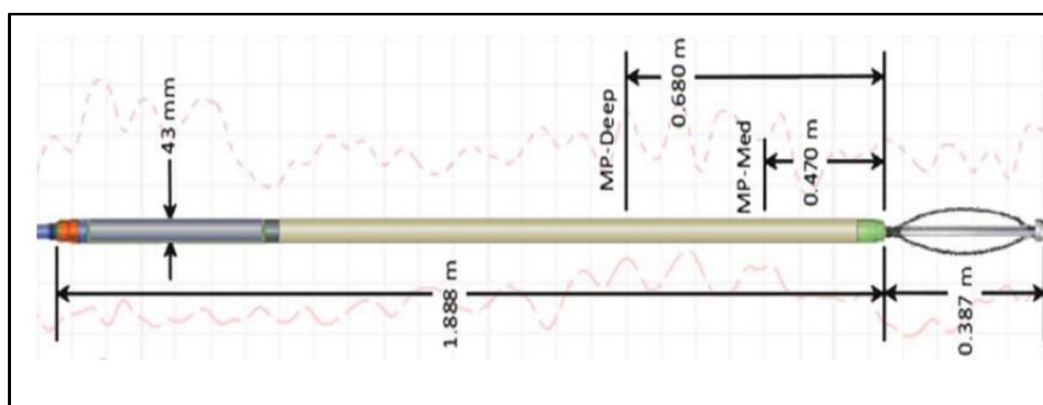


Figure 1—Novel Resistivity Tool used in the studied well

Data Analysis and Verification. Post-retrieval, the recorded Resistivity, Gamma-Ray, and flowmeter data were processed using petrophysical software to calculate saturation profiles and map fluid front movement. Resistivity data were compared against baseline Logging-While-Drilling based logs to assess water saturation changes, correcting for acid-induced effects like saline byproducts or wormholes. Flowmeter was ran however, due to poor spinner response, data could not identify water influx sources. The integrated dataset of resistivity as well as PNL provided a comprehensive assessment of the well's saturation state at present date without the need of a Rig, supporting effective water shut-off planning as well as deployment of artificial lifting techniques in the complex open-hole environment. This methodology ensured the novel resistivity-based, rigless logging solution, operating in memory mode, effectively addressed PNL's limitations, delivering robust saturation data and supporting scalable applications in similar carbonate reservoirs.

Results: Overcoming Pulse Neutron Logging Limitations

The novel resistivity-based, rigless logging solution, deployed in November 2024 via 2-inch coiled tubing in memory mode, revealed a true resistivity decrease from 55 ohm-m to 2 ohm-m that translated into water saturation increase from 13% to 53% in an open-hole carbonate reservoir (Figure 2). These findings explained the well's production cessation due to water loading. In the case of PNL logs, which are basically shallow reading, they are affected by the presence of near-wellbore water, and it shows 79% water saturation in the studied well (Figure 3). Unlike pulse neutron logging² (PNL), which is distorted by saline byproducts, acid-induced wormholes, shallow 6–12-inch investigation, and subjectivity in the log saturation calculation, the studied tool's deep resistivity measurements (24–48 inches) captured unaltered formation properties and unbiased saturation interpretation. The 53% water saturation is within the 50–60% water cut threshold for gas lifting or ESP installation, suggesting the immediate need for the application of gas lift technology. Fortunately, the well was already completed with a side pocket mandrel. Eventually, when it reaches a watered-out state, rig entry to plug and redrill in a new zone would be necessitated. These results empowered the organization to take data-driven intervention planning, avoid unnecessary expenditure on sidetracking, and increase the longevity of the present wellbore with the capability of producing economically, demonstrating the tool's effectiveness in saturation monitoring for complex open-hole reservoirs. As far as limitation is concerned, this tool independently cannot detect crossflows as it needs to be supported by flowmeter readings and must be run in an open-hole condition.

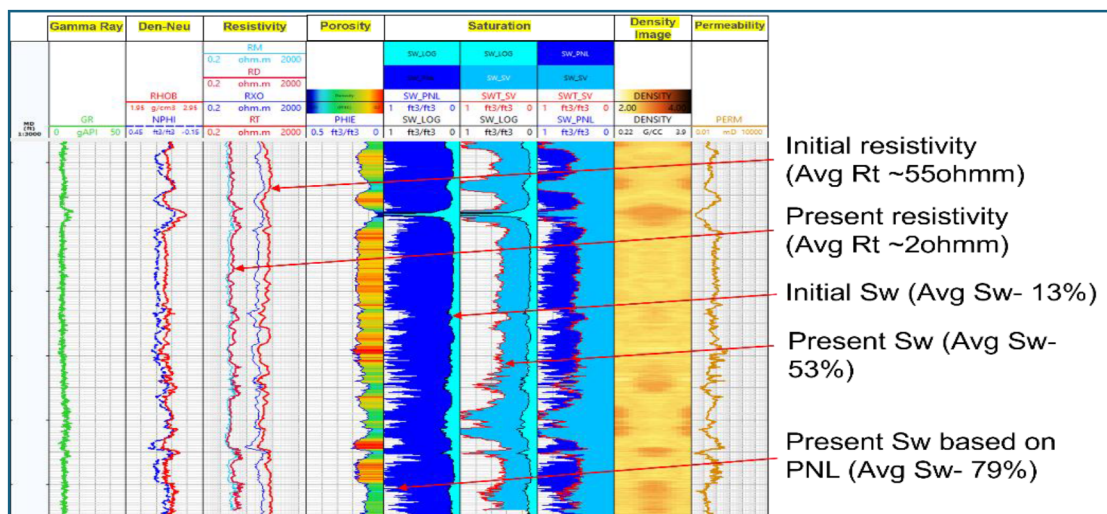


Figure 2—Comparison of Original resistivity versus Riglessly acquired resistivity (saturation tab).

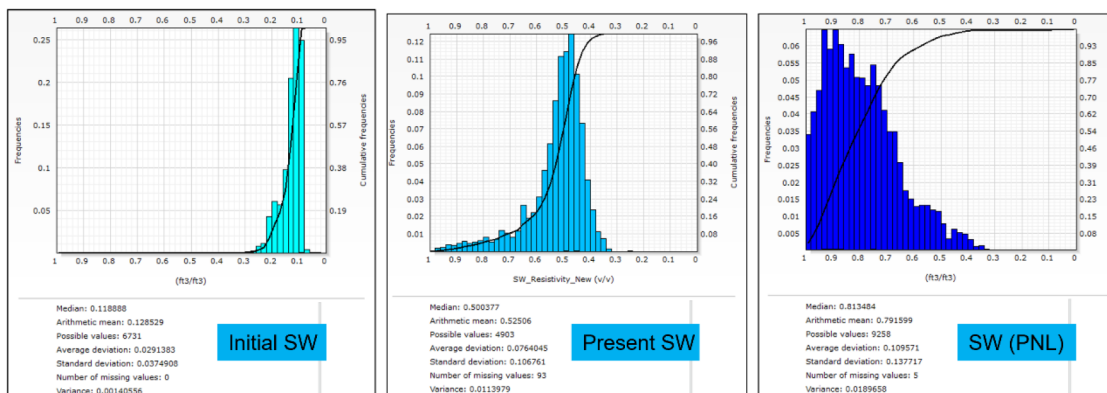


Figure 3—Frequency plot of different water saturation (Initial and current resistivity as well as PNL)

Flip Side: Weighing the Pros and Cons

The novel resistivity-based, rigless logging solution offers substantial advantages over conventional resistivity logs and pulse neutron logging (PNL) for saturation monitoring in open-hole carbonate reservoirs, though its applicability is constrained by a specific limitation. Unlike traditional resistivity logs, which are typically run during drilling and require larger equipment, this sleek, compact tool, deployed via 1.5–2-inch coiled tubing in memory mode, can be run through various sizes of tubing in any open-hole well, enabling precise measurement of current water saturation without rig intervention. This capability proved critical in a recent deployment, where the tool identified a water saturation increase from 13% to 53% and a resistivity decrease from 55 ohm-m to 2 ohm-m, reliably diagnosing water loading in a non-flowing well, a scenario where PNL struggles due to its sensitivity to borehole and acid-induced interferences. The tool's deep resistivity measurements, probing 24–48 inches into the formation, significantly surpass PNL's shallow 6–12-inch investigation depth, capturing unaltered formation properties unaffected by saline byproducts, acid-induced wormholes, or scale precipitation, ensuring robust data in complex reservoirs.

However, the tool's primary limitation is its restriction to open-hole completions, as it cannot be effectively deployed in cased-hole wells where casing and cement introduce signal attenuation, necessitating alternative logging methods. Despite this constraint, the solution's versatility, operational efficiency, and ability to deliver reliable saturation data in challenging open-hole environments, particularly in non-flowing wells with high water cut, make it a transformative tool for optimizing reservoir management in carbonate fields.

The financial impact was highly visible. By delivering repeatable and direct time-lapse, the time lapse resistivity enables operators to take concrete, data-driven decisions on whether to pursue Rigless interventions to restore production or commit to rig-based recompletions in alternate reservoirs.

While rig interventions can also yield useful datasets, they often coincide with limited equipment availability (lower completions accessories, WSO chemicals, etc.), due to no lead time between data acquisition and the need to make a decision on next steps. This often leads to sub-optimal recompletion choices or the need for costly repeat interventions.

Metric	Rigless Time Lapse Resistivity	Pulsed Neutron Logging	Rig-Based Logging & Intervention
Data Confidence	High (direct saturation, open hole)	Moderate (indirect, model dependent)	High (but tied to rig schedule)
Repeatability	Easy, low-cost repeats	Easy, low-cost repeats	Low (rig commitment required)
Decision Quality	Strong, actionable early	Ambiguous in complex reservoirs	Often constrained by equipment timing
Relative Cost	~5% of rig intervention	~5% of rig intervention	100% baseline

Conclusions

- The introduction of the novel resistivity-based, Rigless logging solution represents a transformative leap forward in time-lapse saturation monitoring for open-hole carbonate reservoirs, offering a robust alternative to conventional pulse neutron logging (PNL) and redefining reservoir surveillance practices.
- Deployed via small-diameter coiled tubing in memory mode, this compact, multi-sensor tool integrates deep and medium resistivity measurements, gamma-ray, and flowmeter capabilities to deliver precise, objective data, effectively addressing PNL's limitations, including shallow investigation depth, susceptibility to borehole interferences, and distortions from acid-induced effects in complex, highly deviated environments.

- Its ability to navigate challenging open-hole conditions, such as rugosity and acid-altered zones, while capturing deeper formation properties ensures reliable identification of fluid saturation changes and water influx sources, facilitating targeted reservoir management strategies.
- The rigless, through-tubing deployment enhances operational efficiency, reducing costs and eliminating the need for rig interventions, making it particularly suited for mature fields facing production challenges like water loading.
- This tool independently cannot detect crossflows as it needs to be supported by flowmeter readings and water flow logs.
- While the tool's applicability is limited to open-hole completions, restricting its use in cased-hole wells due to casing and cement effects, its versatility in non-flowing wells and resilience to low-salinity or acid-related interferences establishes it as a game-changer for carbonate reservoir management.
- Future enhancements, such as integrating real-time telemetry or enhancing resistivity contrast, promise to broaden its utility across diverse reservoir types, further optimizing production and surveillance strategies.
- This pioneering approach not only resolves longstanding industry challenges of saturation-production discrepancies but also establishes a scalable, cost-effective framework for advance reservoir management, setting a new standard for petroleum engineering innovation and paving the way for enhanced monitoring and optimization in complex carbonate reservoirs worldwide.

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